

AY 2025-26

Web Technology

Quality Laboratory Manual

Experiment No. 03

Program Based on JavaScript operators, functions and objects



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Experiment No. 03

Title of Experiment: Program Based on Javascript operators, functions and objects

Aim of Experiment: To develop a web page using JavaScript operators, functions, and objects to perform dynamic client-side scripting operations.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment – Web Browser, Text Editor, VSCode

Experiment Objectives:

- Understand JavaScript operators and expressions
- Implement user-defined functions in JavaScript
- Create and use JavaScript objects
- Develop dynamic web pages using client-side scripting

Experiment Outcomes:

- Understand the basics of client-side scripting using JavaScript.
- Apply various JavaScript operators to perform arithmetic and logical operations.
- Develop and use JavaScript functions for structured and reusable code.
- Create and manipulate JavaScript objects for storing and managing data.
- Design interactive and dynamic web pages using JavaScript and user events.

Theory:

JavaScript is a client-side scripting language used to create dynamic and interactive web pages. It runs inside the web browser and allows developers to perform operations such as form validation, calculations, event handling, and dynamic content updates without reloading the page. JavaScript supports programming concepts like operators, functions, and objects which help in building modular and efficient web applications.

1. JavaScript Operators

Operators are symbols used to perform operations on variables and values. They help in performing calculations, comparisons, and logical decisions within programs.

Types of JavaScript Operators:

- **Arithmetic Operators:** Used for mathematical calculations
Example: +, -, *, /, %
- **Comparison Operators:** Used to compare two values
Example: ==, ===, !=, >, <
- **Logical Operators:** Used for conditional operations
Example: && (AND), || (OR), ! (NOT)

- **Assignment Operators:** Used to assign values
Example: = , += , -= , *=

2. JavaScript Functions

Functions are reusable blocks of code designed to perform specific tasks. They improve program structure, reduce code repetition, and enhance maintainability.

Features of Functions:

- Defined using function keyword
- Can accept parameters and return values
- Can be called multiple times
- Help in modular programming

3. JavaScript Objects

Objects are collections of related data and functions stored in key-value pairs. They help in organizing complex data in a structured manner.

Characteristics:

- Store multiple values in one variable
- Represent real-world entities like student, product, employee
- Contain properties and methods

4. Events in JavaScript

Events are actions performed by users such as clicking a button, entering data, or moving the mouse. JavaScript responds to these events using event handlers.

Common Events:

- **OnClick-** The onclick event occurs when a user clicks on an HTML element such as a button, link, or image. It is commonly used to perform actions like calculations, displaying messages, or changing content dynamically.
- **Onchange-** The onchange event is triggered when the value of an input field, dropdown, or textarea changes and loses focus. It is mainly used for validation and updating data automatically when users modify input values.
- **Onload-** The onload event occurs when a web page or element is fully loaded in the browser. It is commonly used to initialize scripts, display welcome messages, or load data automatically when the page opens.
- **Onsubmit-** The onsubmit event is triggered when a form is submitted by the user. It is mainly used for validating form data before sending it to the server.

Program Code:

```
<!DOCTYPE html>
<html>
<head>
<title>JavaScript Operators Functions Objects</title>
<script>
function calculate(){
    var a = parseInt(document.getElementById("num1").value);
    var b = parseInt(document.getElementById("num2").value);

    var result = {
        addition: a + b,
        subtraction: a - b,
        multiplication: a * b,
        division: a / b
    };

    document.getElementById("output").innerHTML =
        "Addition: " + result.addition + "<br>" +
        "Subtraction: " + result.subtraction + "<br>" +
        "Multiplication: " + result.multiplication + "<br>" +
        "Division: " + result.division;
}
</script>
</head>
<body>
<h2>JavaScript Experiment</h2>
Enter Number 1: <input type="number" id="num1"><br><br>
Enter Number 2: <input type="number" id="num2"><br><br>
<button onclick="calculate()">Calculate</button>
<p id="output"></p>
</body>
</html>
```

Task for Students: Design a “Customer Bill Generator” using JavaScript operators, functions and objects

Task Requirements:

- Create an HTML form to accept customer details such as customer name, product name, quantity, and price.
- Use JavaScript operators to perform bill calculations like total amount, tax, and final payable amount.
- Implement JavaScript functions to calculate total price and generate the bill output.
- Use JavaScript objects to store customer information and billing details.
- Apply event handling (onclick / onsubmit) to trigger bill generation.
- Display the generated bill dynamically using DOM manipulation (Dynamic HTML)

Submission Requirement:

- 1) customer_bill_generator.html (including HTML + JavaScript code)
- 2) Screenshots of output

Observations:

- JavaScript operators were successfully used to perform arithmetic calculations such as total amount, tax, and final bill generation.
- Functions helped in organizing reusable code and simplified the billing calculation process.
- JavaScript objects were effectively used to store customer details and billing information in a structured format.
- Event handling enabled dynamic interaction between the user and the web page during bill generation.
- Dynamic HTML and DOM manipulation allowed real-time display of the generated customer bill without refreshing the page.

Conclusion:

This experiment demonstrated the practical use of JavaScript operators, functions, and objects in developing a dynamic Customer Bill Generator. Operators were used to perform billing calculations efficiently, while functions helped in organizing reusable program logic. JavaScript objects enabled structured storage of customer and product details. Event handling and Dynamic HTML improved user interaction by generating the bill instantly without reloading the page.

Expected Oral Questions:

1. What are JavaScript operators?
2. Difference between function and object?
3. What is DOM manipulation?
4. Explain event handling in JavaScript?

FAQs in Interview:

1. What is client-side scripting in JavaScript?
2. What are JavaScript operators? Explain types.
3. What is a function in JavaScript?
4. Difference between function and object in JavaScript.
5. What is a JavaScript object?
6. What is event handling in JavaScript?
7. What is DOM manipulation?
8. What is Dynamic HTML (DHTML)?
9. Difference between == and === operator.
10. How does JavaScript improve user interaction in web applications?

